Task History

Initiating Search

November 10, 2025, 11:14 AM

Search:

CAS Lexicon Search:

Wolff-Kishner reaction - Preferred Concept

Search Tasks

Task	Result Type	View
Returned Reference Results from CAS Lexicon (394)	References	View Results
Exported: Retrieved Related Reaction Results + Filters (407) Filtered By: Document Type: **Journal**	Reactions	View Results

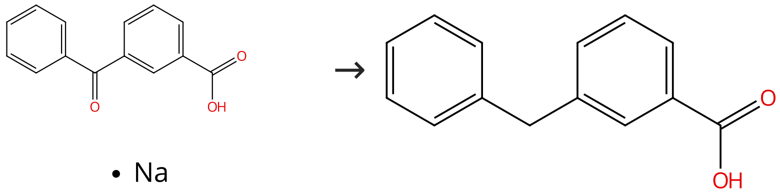
CAS SciFinder

Reactions (50)

View in CAS SciFinder

Scheme 1 (1 Reaction)

Steps: **1** Yield: **100%**



Supplier (1)

Suppliers (38)

31-614-CAS-33997565

Steps: **1** Yield: **100%**

1.1 **Reagents:** Potassium *tert*-butoxide, Hydrazine hydrate (1:1)
Catalysts: Dimethyl sulfoxide, (*OC*-6-15)-Dichloro[*N*-[2-(diphenylphosphino-κ*P*)ethyl]-2-pyridinemethanamine-κN^1, κN^2](triphenylphosphine)ruthenium
Solvents: Water; overnight, 75 °C

Experimental Protocols

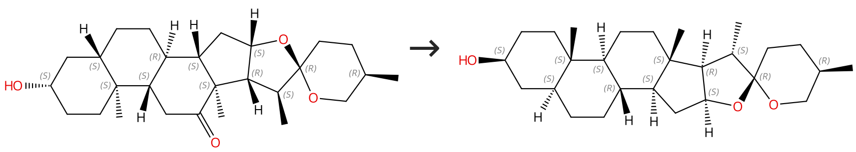
Ruthenium(II)-catalyzed deoxygenation of ketones

By: Gui, Ruohua; et al

Chemical Communications (Cambridge, United Kingdom) (2022), 58(75), 10572-10575.

Scheme 2 (1 Reaction)

Steps: **1** Yield: **100%**



Absolute stereochemistry shown

Absolute stereochemistry shown

Suppliers (64)

Suppliers (64)

31-538-CAS-8909179

Steps: **1** Yield: **100%**

1.1 **Reagents:** Hydrazine hydrate (1:1)
Solvents: Glycol monoethyl ether

1.2 **Reagents:** Potassium hydroxide

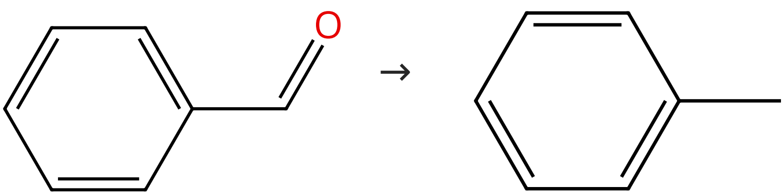
Optimization of the Wolff-Kishner reaction for the obtainment of tigogenin from hecogenin

By: Moreno, Mayra Reyes; et al

Revista CENIC, Ciencias Quimicas (2001), 32(1), 51-54.

Scheme 3 (2 Reactions)

Steps: **1** Yield: **92-100%**



Suppliers (69)

Suppliers (280)

31-538-CAS-5254410 Steps: **1** Yield: **100%**

1.1 **Reagents:** Hydrazine
Solvents: Glycerol; 5 min, rt → 90 °C

1.2 **Reagents:** Potassium hydroxide; 5 min, heated

Glycerol as an alternative green medium for carbonyl compound reductions

By: Wolfson, Adi; et al

Organic Communications (2009), 2(2), 34-41.

31-538-CAS-15006501 Steps: **1** Yield: **92%**

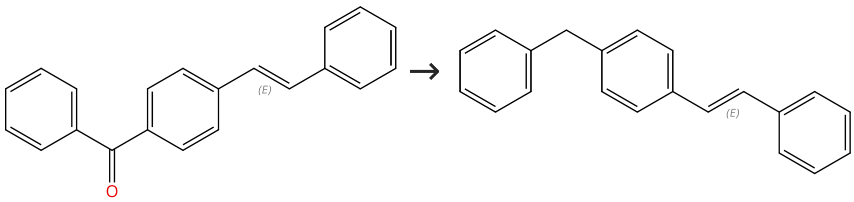
1.1 **Reagents:** Hydrazine hydrate (1:1)
Catalysts: Amberlyst A 26
Solvents: Ethylene glycol; 55 min, reflux

An efficient and environmentally friendly Wolff-Kishner reduction catalyzed by strong base anion-exchange resin

By: More, Uttam B.; et al

Organic Chemistry: An Indian Journal (2008), 4(12), 537-539.

Scheme 4 (1 Reaction) Steps: **1** Yield: **99%**



Double bond geometry shown

Double bond geometry shown

Suppliers (5)

Suppliers (5)

31-614-CAS-33997553 Steps: **1** Yield: **99%**

1.1 **Reagents:** Potassium *tert*-butoxide, Hydrazine hydrate (1:1)
Catalysts: Dimethyl sulfoxide, (*OC*-6-15)-Dichloro[*N*-[2-(diphenylphosphino-κ*P*)ethyl]-2-pyridinemethanamine-κN^1, κN^2](triphenylphosphine)ruthenium
Solvents: Methanol; overnight, 75 °C

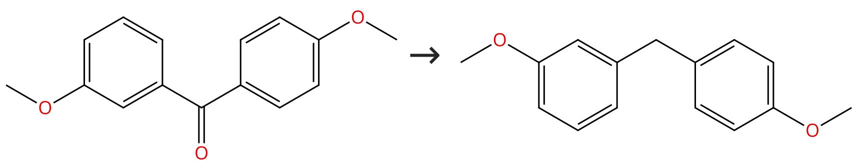
Experimental Protocols

Ruthenium(II)-catalyzed deoxygenation of ketones

By: Gui, Ruohua; et al

Chemical Communications (Cambridge, United Kingdom) (2022), 58(75), 10572-10575.

Scheme 5 (1 Reaction) Steps: **1** Yield: **99%**



Suppliers (33)

Suppliers (5)

31-614-CAS-33997552 Steps: **1** Yield: **99%**

1.1 **Reagents:** Potassium *tert*-butoxide, Hydrazine hydrate (1:1)
Catalysts: Dimethyl sulfoxide, (*OC*-6-15)-Dichloro[*N*-[2-(diphenylphosphino-κ*P*)ethyl]-2-pyridinemethanamine-κN^1, κN^2](triphenylphosphine)ruthenium
Solvents: Methanol; overnight, 75 °C

Experimental Protocols

Ruthenium(II)-catalyzed deoxygenation of ketones

By: Gui, Ruohua; et al

Chemical Communications (Cambridge, United Kingdom) (2022), 58(75), 10572-10575.

Scheme 6 (1 Reaction) Steps: **1** Yield: **99%**

31-614-CAS-33997548 Steps: **1** Yield: **99%**

1.1 **Reagents:** Potassium *tert*-butoxide, Hydrazine hydrate (1:1)
Catalysts: Dimethyl sulfoxide, (*OC*-6-15)-Dichloro[*N*-[2-(diphenylphosphino-κ*P*)ethyl]-2-pyridinemethanamine-κN^1, κN^2](triphenylphosphine)ruthenium
Solvents: Methanol; overnight, 75 °C

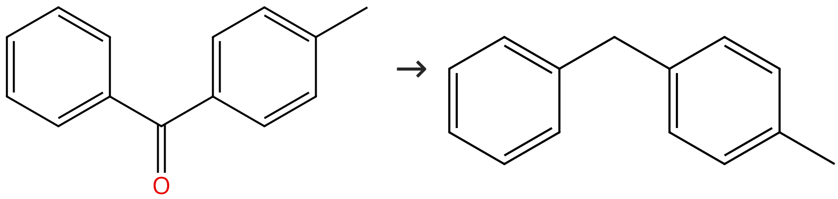
Experimental Protocols

Ruthenium(II)-catalyzed deoxygenation of ketones

By: Gui, Ruohua; et al

Chemical Communications (Cambridge, United Kingdom) (2022), 58(75), 10572-10575.

Scheme 7 (1 Reaction) Steps: **1** Yield: **99%**



Suppliers (81) Suppliers (61)

31-614-CAS-33997551 Steps: **1** Yield: **99%**

1.1 **Reagents:** Potassium *tert*-butoxide, Hydrazine hydrate (1:1)
Catalysts: Dimethyl sulfoxide, (*OC*-6-15)-Dichloro[*N*-[2-(diphenylphosphino-κ*P*)ethyl]-2-pyridinemethanamine-κN^1, κN^2](triphenylphosphine)ruthenium
Solvents: Methanol; overnight, 75 °C

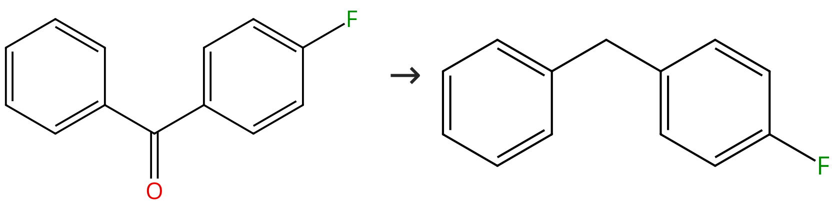
Experimental Protocols

Ruthenium(II)-catalyzed deoxygenation of ketones

By: Gui, Ruohua; et al

Chemical Communications (Cambridge, United Kingdom) (2022), 58(75), 10572-10575.

Scheme 8 (1 Reaction) Steps: **1** Yield: **99%**



Suppliers (71) Suppliers (48)

31-614-CAS-33997563 Steps: **1** Yield: **99%**

1.1 **Reagents:** Potassium *tert*-butoxide, Hydrazine hydrate (1:1)
Catalysts: Dimethyl sulfoxide, (*OC*-6-15)-Dichloro[*N*-[2-(diphenylphosphino-κ*P*)ethyl]-2-pyridinemethanamine-κN^1, κN^2](triphenylphosphine)ruthenium
Solvents: Water; overnight, 75 °C

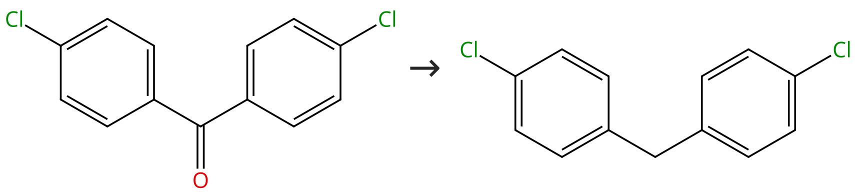
Experimental Protocols

Ruthenium(II)-catalyzed deoxygenation of ketones

By: Gui, Ruohua; et al

Chemical Communications (Cambridge, United Kingdom) (2022), 58(75), 10572-10575.

Scheme 9 (1 Reaction) Steps: **1** Yield: **99%**



Suppliers (80) Suppliers (32)

31-614-CAS-33997559 Steps: **1** Yield: **99%**

1.1 **Reagents:** Potassium *tert*-butoxide, Hydrazine hydrate (1:1)
Catalysts: Dimethyl sulfoxide, (*OC*-6-15)-Dichloro[*N*-[2-(diphenylphosphino-κ*P*)ethyl]-2-pyridinemethanamine-κN^1, κN^2](triphenylphosphine)ruthenium
Solvents: Water; overnight, 75 °C

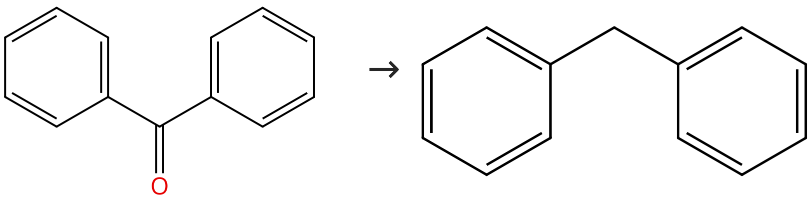
Experimental Protocols

Ruthenium(II)-catalyzed deoxygenation of ketones

By: Gui, Ruohua; et al

Chemical Communications (Cambridge, United Kingdom) (2022), 58(75), 10572-10575.

Scheme 10 (1 Reaction) Steps: **1** Yield: **99%**



Suppliers (136) Suppliers (86)

31-614-CAS-33997549 Steps: **1** Yield: **99%**

1.1 **Reagents:** Potassium *tert*-butoxide, Hydrazine hydrate (1:1)
Catalysts: Dimethyl sulfoxide, (*OC*-6-15)-Dichloro[*N*-[2-(diphenylphosphino-κ*P*)ethyl]-2-pyridinemethanamine-κN^1, κN^2](triphenylphosphine)ruthenium
Solvents: Methanol; overnight, 75 °C

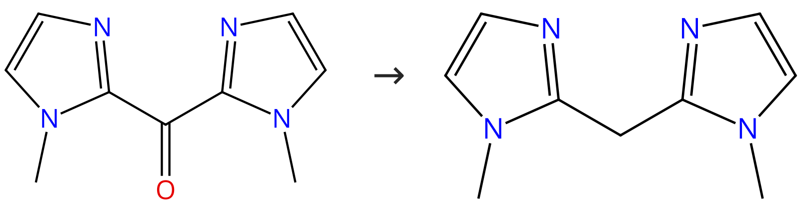
Experimental Protocols

Ruthenium(II)-catalyzed deoxygenation of ketones

By: Gui, Ruohua; et al

Chemical Communications (Cambridge, United Kingdom) (2022), 58(75), 10572-10575.

Scheme 11 (1 Reaction) Steps: **1** Yield: **98%**



Suppliers (35) Suppliers (39)

31-538-CAS-17259383 Steps: **1** Yield: **98%**

1.1 **Reagents:** Potassium hydroxide, Hydrazine hydrate (1:1)
Solvents: Water; 3 h, 150 °C

Experimental Protocols

From Bis(imidazol-2-yl)methanes to Asymmetrically Substituted Bis(heterocyclo)methanides in Metal Coordination

By: Dauer, David-R.; et al

European Journal of Inorganic Chemistry (2017), 2017(13), 1966-1978.

Scheme 12 (3 Reactions) Steps: **1** Yield: **98%**

Suppliers (85) Suppliers (75)

31-538-CAS-14642882	Steps: **1** Yield: **98%**
1.1	**Reagents:** Potassium hydroxide, Hydrazine hydrate (1:1) **Solvents:** 2-Methoxyethanol, Water; 2 h, 70 °C; 1 h, 120 °C; 2 h, 150 °C

Alternative synthesis of 5-chloromethyl-2,3-dimethoxy-6-methyl-1,4-benzoquinone: a key intermediate for preparing coenzyme Q analogues

By: Wang, Jin; et al

Journal of Chemical Research (2010), 34(12), 724-725.

31-538-CAS-6630263	Steps: **1** Yield: **98%**
1.1	**Reagents:** Potassium hydroxide, Hydrazine hydrate (1:1) **Solvents:** Ethylene glycol; 2 h, 70 °C; 1 h, 120 °C; 2 h, 150 °C

A green and efficient synthesis of 1-chloromethyl-2,3,4,5-tetramethoxy-6-methylbenzene

By: Wang, Jin; et al

Journal of Chemical Research (2010), 34(12), 717-718.

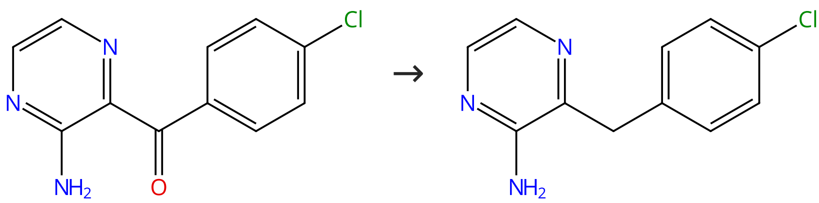
31-538-CAS-2568319	Steps: **1** Yield: **98%**
1.1	**Reagents:** Hydrazine hydrate (1:1) **Solvents:** Ethylene glycol, Water; rt → 70 °C
1.2	**Reagents:** Potassium hydroxide; 3 h, 70 °C; 70 °C → 125 °C; 1.5 h, 125 °C; 125 °C → 150 °C; 3 h, 150 °C

Asymmetric synthesis of piperazine derivative

By: Sun, Jiaqiang; et al

Xiandai Huagong (2009), 29(11), 59-60.

Scheme 13 (1 Reaction) Steps: **1** Yield: **97%**



Suppliers (4) Suppliers (3)

31-538-CAS-787016	Steps: **1** Yield: **97%**
1.1	**Reagents:** Hydrazine hydrate (1:1) **Solvents:** Ethylene glycol; 1 h, 100 °C; 100 °C → rt
1.2	**Reagents:** Potassium hydroxide; 30 min, 240 °C

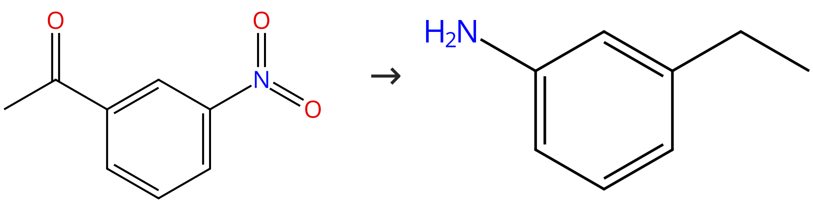
Experimental Protocols

New red-shifted coelenterazine analogues with an extended electronic conjugation

By: Giuliani, Germano; et al

Tetrahedron Letters (2012), 53(38), 5114-5118.

Scheme 14 (1 Reaction) Steps: **1** Yield: **97%**



Suppliers (84) Suppliers (80)

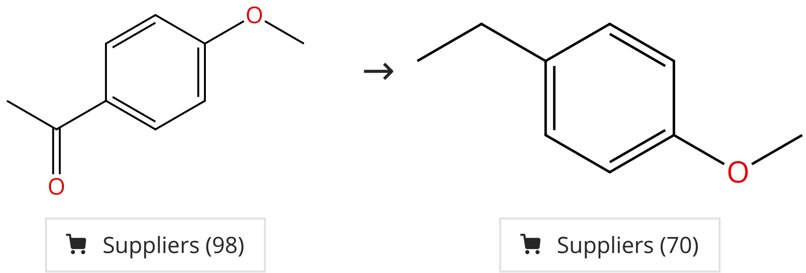
31-522-CAS-7399148	Steps: **1** Yield: **97%**
1.1 **Reagents:** Hydrazine, Potassium hydroxide; 24 h, 135 °C	
Experimental Protocols	

One-step double reduction of aryl nitro and carbonyl groups using hydrazine

By: Diez-Cecilia, Elena; et al

Tetrahedron Letters (2011), 52(50), 6702-6704.

Scheme 15 (1 Reaction) Steps: **1** Yield: **97%**



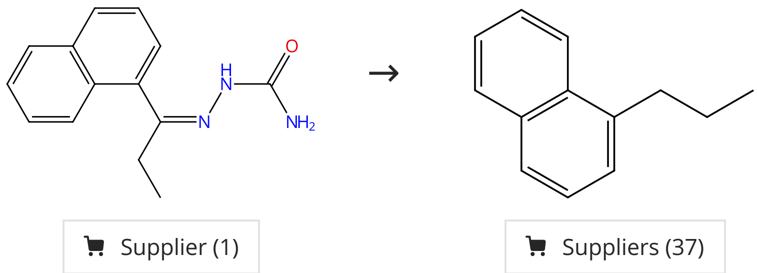
31-538-CAS-5275280	Steps: **1** Yield: **97%**
1.1 **Reagents:** Hydrazine	
1.2 **Reagents:** Potassium hydroxide	
Experimental Protocols	

Microwave induced synthesis of hydrazones and Wolff-Kishner reduction of carbonyl compounds

By: Gadhwal, Sunil; et al

Synlett (1999), (10), 1573-1574.

Scheme 16 (1 Reaction) Steps: **1** Yield: **96%**



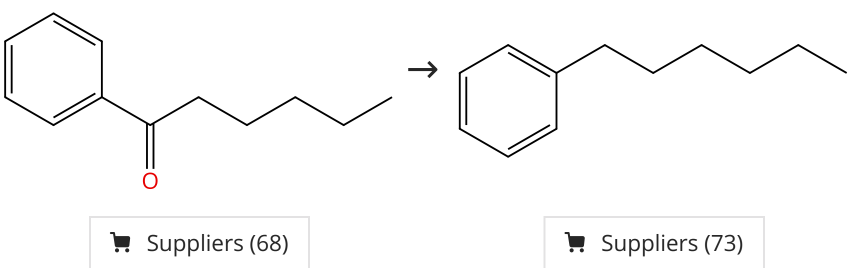
31-538-CAS-21361448	Steps: **1** Yield: **96%**
1.1 **Reagents:** Potassium hydroxide **Solvents:** Diethylene glycol; 4.5 h, reflux	

Synthesis of n-alkylnaphthalenes via semicarbazones

By: Zengin, Gulay; et al

Turkish Journal of Chemistry (2006), 30(2), 139-144.

Scheme 17 (1 Reaction) Steps: **1** Yield: **96%**



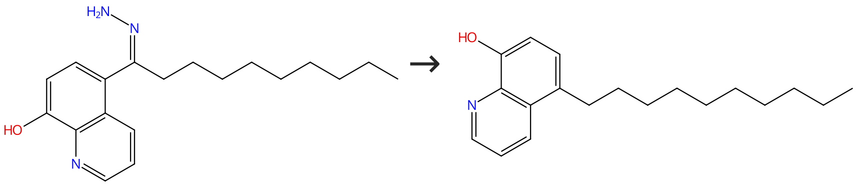
31-538-CAS-13218023	Steps: **1** Yield: **96%**
1.1 **Reagents:** Potassium hydroxide, Hydrazine hydrate (1:1) **Solvents:** Diethylene glycol monoethyl ether; 10 min, 200 psi, 220 °C	
Experimental Protocols	

Rapid Wolff-Kishner reductions in a silicon carbide microreactor

By: Newman, Stephen G.; et al

Green Chemistry (2014), 16(1), 176-180.

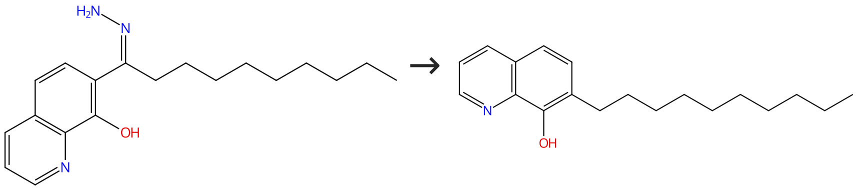
Scheme 18 (1 Reaction) Steps: **1** Yield: **95%**



Supplier (1) Suppliers (5)

31-538-CAS-21433158 Steps: **1** Yield: **95%**	**Chelates of quinolin-8-ol derivatives. II. Synthesis of long-chain n-alkylquinolin-8-ols**
No Data Available	By: Uhlemann, Erhard; et al Zeitschrift fuer Chemie (1983), 23(9), 334.

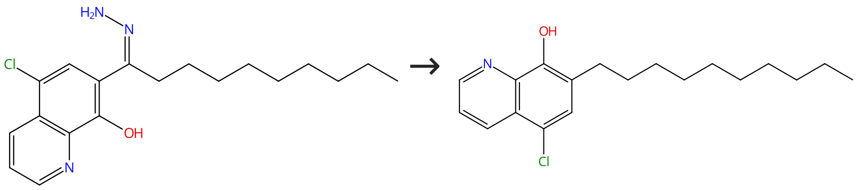
Scheme 19 (1 Reaction) Steps: **1** Yield: **95%**



Suppliers (4) Suppliers (4)

31-538-CAS-21432443 Steps: **1** Yield: **95%**	**Chelates of quinolin-8-ol derivatives. II. Synthesis of long-chain n-alkylquinolin-8-ols**
No Data Available	By: Uhlemann, Erhard; et al Zeitschrift fuer Chemie (1983), 23(9), 334.

Scheme 20 (1 Reaction) Steps: **1** Yield: **95%**

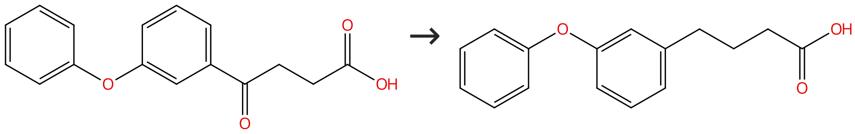


Suppliers (4) Suppliers (4)

31-538-CAS-21430534 Steps: **1** Yield: **95%**	**Chelates of quinolin-8-ol derivatives. II. Synthesis of long-chain n-alkylquinolin-8-ols**
No Data Available	By: Uhlemann, Erhard; et al Zeitschrift fuer Chemie (1983), 23(9), 334.

Scheme 21 (1 Reaction)

Steps: **1** Yield: **95%**



Suppliers (29) Suppliers (4)

31-538-CAS-5623854 Steps: **1** Yield: **95%**

1.1 **Reagents:** Hydrazine hydrate (1:1)
Solvents: Triethylene glycol

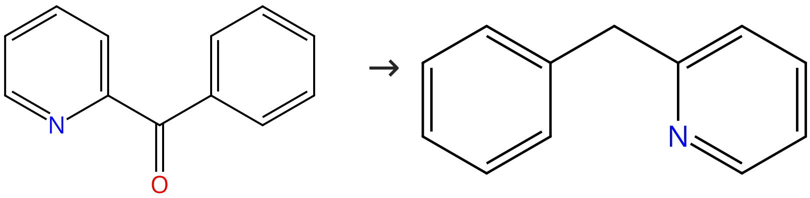
Simple modification of the Wolff-Kishner reduction

By: Huang, Minlon

Journal of the American Chemical Society (1946), 68, 2487-8.

Scheme 22 (1 Reaction)

Steps: **1** Yield: **95%**



Suppliers (97) Suppliers (64)

31-614-CAS-33997569 Steps: **1** Yield: **95%**

1.1 **Reagents:** Potassium *tert*-butoxide, Hydrazine hydrate (1:1)
Catalysts: Dimethyl sulfoxide, (*OC*-6-15)-Dichloro[*N*-[2-(diphenylphosphino-κ*P*)ethyl]-2-pyridinemethanamine-κ*N*[1], κ*N*[2]](triphenylphosphine)ruthenium
Solvents: Methanol; overnight, 75 °C

Experimental Protocols

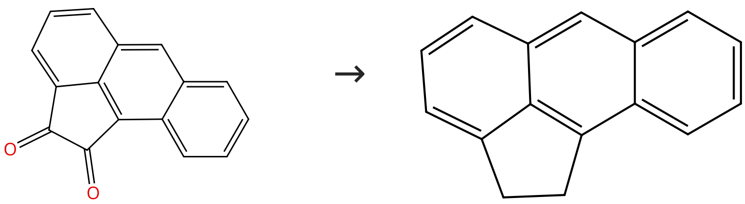
Ruthenium(II)-catalyzed deoxygenation of ketones

By: Gui, Ruohua; et al

Chemical Communications (Cambridge, United Kingdom) (2022), 58(75), 10572-10575.

Scheme 23 (1 Reaction)

Steps: **1** Yield: **95%**



Suppliers (34) Suppliers (8)

31-538-CAS-11660589 Steps: **1** Yield: **95%**

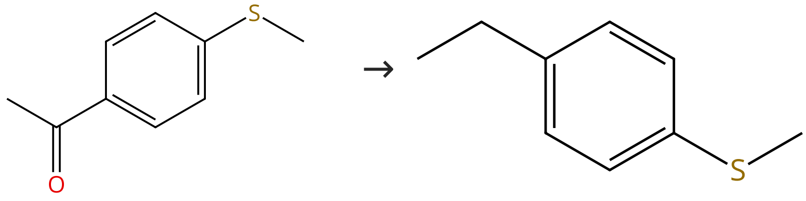
1.1 **Reagents:** Hydrazine
Solvents: Diethylene glycol

The use of a double Wolff- Kishner reduction in the preparation of aceanthrene and aceanthrylene

By: Boerrigter, Jos C. Olde; et al

Recueil des Travaux Chimiques des Pays-Bas (1989), 108(2), 79-80.

Scheme 24 (1 Reaction) Steps: **1** Yield: **94%**



Suppliers (83) Suppliers (24)

31-614-CAS-33997542 Steps: **1** Yield: **94%**

1.1 **Reagents:** Potassium *tert*-butoxide, Hydrazine hydrate (1:1)
Catalysts: Dimethyl sulfoxide, (*OC*-6-15)-Dichloro[*N*-[2-(diphenylphosphino-κ*P*)ethyl]-2-pyridinemethanamine-κN^1, κN^2](triphenylphosphine)ruthenium
Solvents: Methanol; overnight, 75 °C

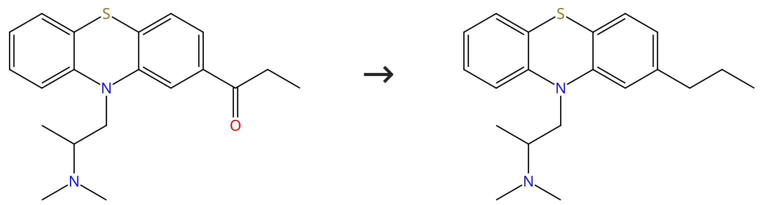
Experimental Protocols

Ruthenium(II)-catalyzed deoxygenation of ketones

By: Gui, Ruohua; et al

Chemical Communications (Cambridge, United Kingdom) (2022), 58(75), 10572-10575.

Scheme 25 (1 Reaction) Steps: **1** Yield: **94%**



Suppliers (33) Supplier (1)

31-538-CAS-14159829 Steps: **1** Yield: **94%**

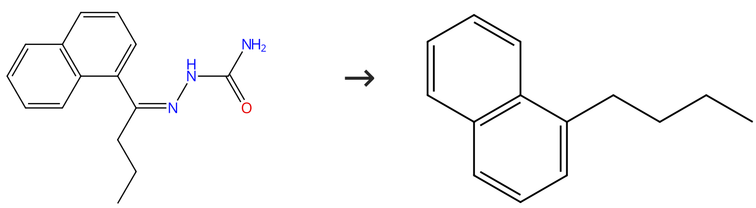
1.1 **Reagents:** Hydrazine
Solvents: Ethanol

Comparative study of the reduction of 2-acyl-10-dialkylaminoalkylphenothiazines

By: Schmitt, Joseph; et al

Bulletin de la Societe Chimique de France (1961), 340-9.

Scheme 26 (1 Reaction) Steps: **1** Yield: **94%**



Supplier (1) Suppliers (49)

31-538-CAS-21367123 Steps: **1** Yield: **94%**

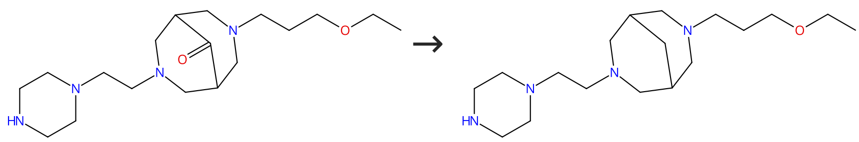
1.1 **Reagents:** Potassium hydroxide
Solvents: Diethylene glycol; 18 h, reflux

Synthesis of n-alkylnaphthalenes via semicarbazones

By: Zengin, Gulay; et al

Turkish Journal of Chemistry (2006), 30(2), 139-144.

Scheme 27 (1 Reaction) Steps: **1** Yield: **94%**



31-538-CAS-16716915 Steps: **1** Yield: **94%**

1.1 **Reagents:** Hydrazine hydrate (1:1)
Solvents: Triethylene glycol

1.2 **Reagents:** Potassium hydroxide; 60 °C; 5 h, 160 - 170 °C; 190 - 200 °C; 1 h, 200 °C → rt

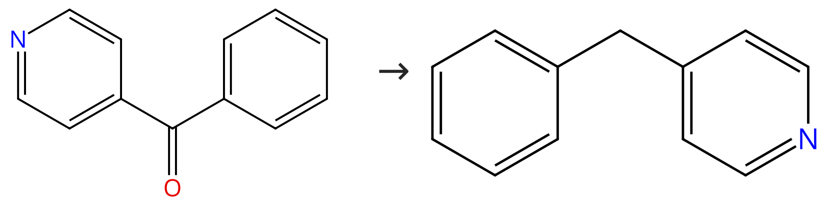
1.3 **Reagents:** Water; cooled

Synthesis of novel 3,7-diazabicyclo[3.3.1]nonane derivatives

By: Malmakova, A. Ye.; et al

Eurasian Chemico-Technological Journal (2014), 16(1), 85-89.

Scheme 28 (1 Reaction) Steps: **1** Yield: **94%**



Suppliers (68) Suppliers (59)

31-538-CAS-2630198 Steps: **1** Yield: **94%**

1.1 **Reagents:** Potassium hydroxide, Hydrazine hydrate (1:1)
Solvents: Diethylene glycol monoethyl ether; 3 min, 200 psi, 200 °C

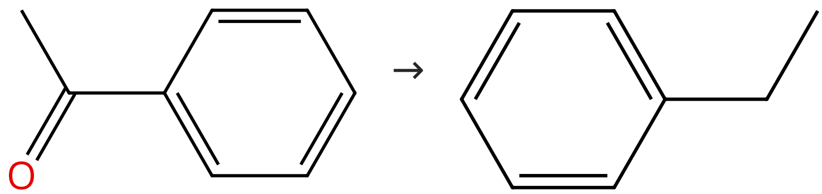
Experimental Protocols

Rapid Wolff-Kishner reductions in a silicon carbide microreactor

By: Newman, Stephen G.; et al

Green Chemistry (2014), 16(1), 176-180.

Scheme 29 (1 Reaction) Steps: **1** Yield: **94%**



Suppliers (105) Suppliers (133)

31-538-CAS-7141870 Steps: **1** Yield: **94%**

1.1 **Reagents:** Hydrazine

1.2 **Reagents:** Potassium hydroxide

Experimental Protocols

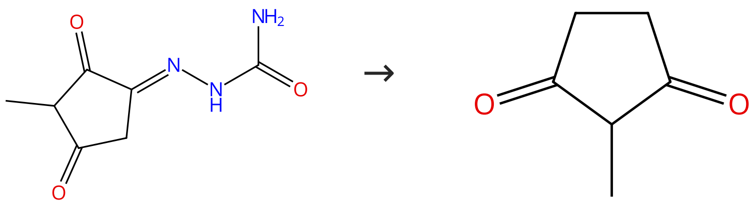
Microwave induced synthesis of hydrazones and Wolff-Kishner reduction of carbonyl compounds

By: Gadhwal, Sunil; et al

Synlett (1999), (10), 1573-1574.

Scheme 30 (1 Reaction)

Steps: **1** Yield: **93%**



Suppliers (4)

Suppliers (79)

31-538-CAS-21226089

Steps: **1** Yield: **93%**

1.1 **Solvents:** Ethylene glycol

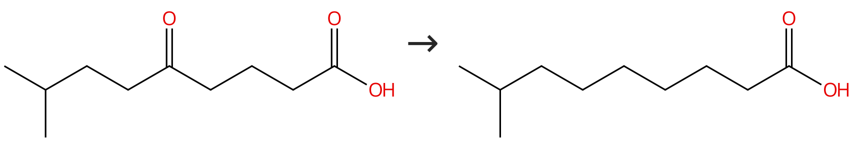
Preparation of 2-methyl-1,3-cyclopentanedione

By: Panouse, Jacques J.; et al

Bulletin de la Societe Chimique de France (1955), 1036-9.

Scheme 31 (1 Reaction)

Steps: **1** Yield: **92%**



Suppliers (4)

Suppliers (62)

31-538-CAS-9139228

Steps: **1** Yield: **92%**

1.1 **Reagents:** Sodium, Hydrazine hydrate (1:1)
Solvents: Diethylene glycol

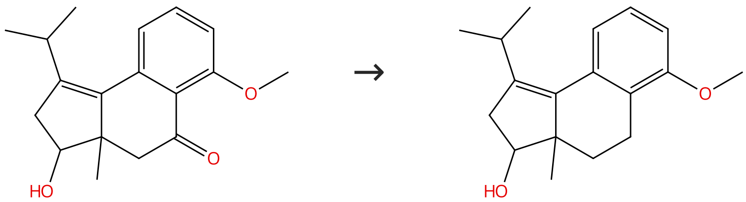
Low-pressure method for Wolff-Kishner reduction

By: Soffer, Milton D.; et al

Journal of the American Chemical Society (1945), 67, 1435-6.

Scheme 32 (1 Reaction)

Steps: **1** Yield: **92%**



31-538-CAS-19883682

Steps: **1** Yield: **92%**

1.1 **Reagents:** Trifluoroacetic acid, Triethylsilane
Solvents: Dichloromethane; 0 °C; 2 h, rt

1.2 **Reagents:** Sodium bicarbonate
Solvents: Water

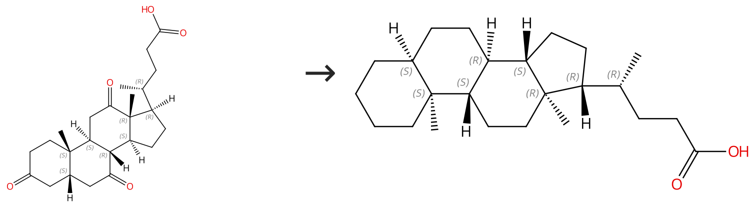
Experimental Protocols

Synthetic Studies on Enantioselective Total Synthesis of Cyathane Diterpenoids: Cyrneines A and B, Glaucopine C, and (+)-Allocyathin B_2

By: Wu, Guo-Jie; et al

Journal of Organic Chemistry (2019), 84(6), 3223-3238.

Scheme 33 (3 Reactions) Steps: **1** Yield: **91-92%**



Absolute stereochemistry shown, Rotation (+)

Suppliers (77)

Absolute stereochemistry shown

Suppliers (41)

31-538-CAS-759276 Steps: **1** Yield: **92%**

1.1 **Reagents:** Hydrazine
Solvents: Diethylene glycol

Reduction of steroid ketones and other carbonyl compounds by modified Wolff-Kishner method

By: Huang-Minlon

Journal of the American Chemical Society (1949), 71, 3301-3.

31-538-CAS-2875361 Steps: **1** Yield: **91%**

1.1 **Reagents:** Hydrazine hydrate (1:1)
Solvents: Diethylene glycol

Reduction of steroid ketones and other carbonyl compounds by modified Wolff-Kishner method

By: Huang-Minlon

Journal of the American Chemical Society (1949), 71, 3301-3.

31-538-CAS-14157605 Steps: **1** Yield: **91%**

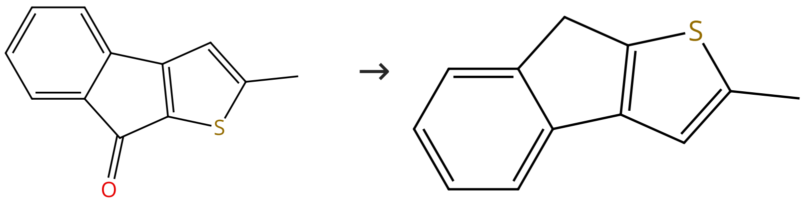
1.1 **Reagents:** Hydrazine hydrate (1:1)
Solvents: Diethylene glycol

Reduction of steroid ketones and other carbonyl compounds by modified Wolff-Kishner method

By: Huang-Minlon

Journal of the American Chemical Society (1949), 71, 3301-3.

Scheme 34 (1 Reaction) Steps: **1** Yield: **92%**



Suppliers (2)

Suppliers (2)

31-538-CAS-9564725 Steps: **1** Yield: **92%**

1.1 **Reagents:** Hydrazine
Solvents: Diethylene glycol; 1 h, 80 °C; 1 h, reflux; reflux → rt

1.2 **Reagents:** Potassium hydroxide
Solvents: Water; rt; 2 h, reflux

Experimental Protocols

Efficient Method for the Synthesis of Hetarenoi ndanones Based on 3-Arylhetarenes and Their Conversion into Hetarenoindenes

By: Kashulin, Igor A.; et al

Journal of Organic Chemistry (2004), 69(16), 5476-5479.

Scheme 35 (1 Reaction) Steps: **1** Yield: **92%**

Suppliers (95) Suppliers (81)

31-538-CAS-4148621 Steps: **1** Yield: **92%**

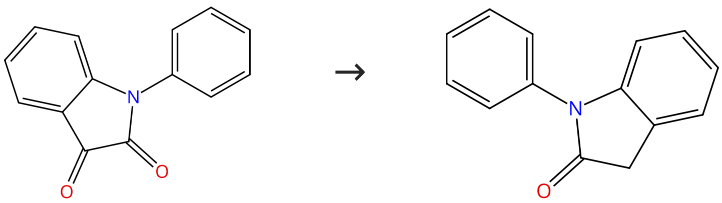
1.1 **Reagents:** Hydrazine

The simplified Wolff-Kishner reaction with aromatic hydroxy carbonyl compounds

By: Lock, G.

Monatshefte fuer Chemie (1954), 85, 802-6.

Scheme 36 (1 Reaction) Steps: **1** Yield: **92%**



Suppliers (67) Suppliers (61)

31-538-CAS-10451801 Steps: **1** Yield: **92%**

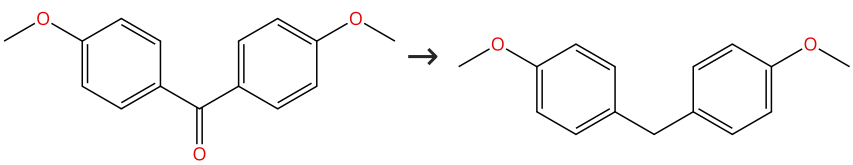
1.1 **Reagents:** Hydrazine hydrate (1:1)

A new efficient and mild synthesis of 2- oxindoles by one-pot Wolff-Kishner-like reduction of isatin derivatives

By: Crestini, C.; et al

Synthetic Communications (1994), 24(20), 2835-41.

Scheme 37 (1 Reaction) Steps: **1** Yield: **92%**



Suppliers (88) Suppliers (46)

31-538-CAS-9547096 Steps: **1** Yield: **92%**

1.1 **Reagents:** Hydrazine

1.2 **Reagents:** Potassium hydroxide

Experimental Protocols

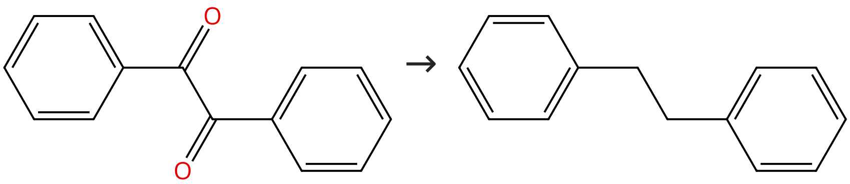
Microwave induced synthesis of hydrazones and Wolff-Kishner reduction of carbonyl compounds

By: Gadhwal, Sunil; et al

Synlett (1999), (10), 1573-1574.

Scheme 38 (1 Reaction)

Steps: **1** Yield: **92%**



Suppliers (101)

Suppliers (86)

31-538-CAS-31834

Steps: **1** Yield: **92%**

1.1 **Reagents:** Hydrazine hydrate (1:1)
Solvents: Ethylene glycol

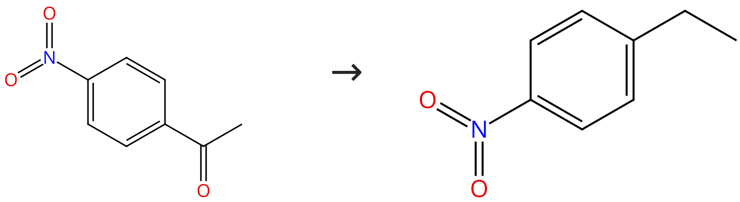
Modified Wolff-Kishner method

By: Huang, Min-Lon; et al

Hua Hseuh Hseuh Pao (1961), 27(1), 1-9.

Scheme 39 (1 Reaction)

Steps: **1** Yield: **92%**



Suppliers (79)

Suppliers (39)

31-538-CAS-7152085

Steps: **1** Yield: **92%**

1.1 **Reagents:** Hydrazine hydrate (1:1)
Catalysts: Amberlyst A 26
Solvents: Ethylene glycol; 55 min, reflux

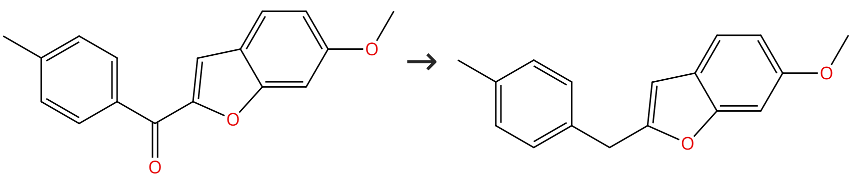
An efficient and environmentally friendly Wolff-Kishner reduction catalyzed by strong base anion-exchange resin

By: More, Uttam B.; et al

Organic Chemistry: An Indian Journal (2008), 4(12), 537-539.

Scheme 40 (1 Reaction)

Steps: **1** Yield: **91%**



Suppliers (4)

Supplier (1)

31-538-CAS-7968309

Steps: **1** Yield: **91%**

1.1 **Reagents:** Hydrazine
Solvents: Ethylene glycol; 15 min, 150 °C

1.2 **Reagents:** Potassium hydroxide; rt; 4 h, rt → 150 °C

1.3 **Reagents:** Hydrochloric acid
Solvents: Water; neutralized

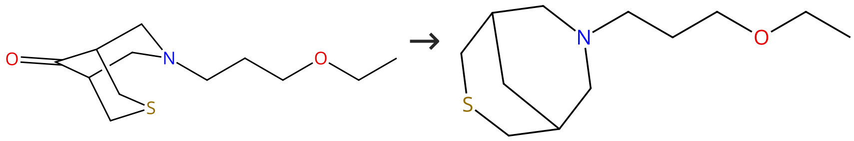
A New Synthetic Route of 2-Aroyl- and 2-Benzyl-Benzofurans and their Application in the Total Synthesis of a Metabolite Isolated from Dorstenia gigas

By: Correa, Christian; et al

Australian Journal of Chemistry (2008), 61(12), 991-999.

Scheme 41 (1 Reaction)

Steps: **1** Yield: **91%**



Supplier (1)

31-538-CAS-4671377

Steps: **1** Yield: **91%**

1.1 **Reagents:** Potassium hydroxide, Hydrazine hydrate (1:1)
Solvents: Triethylene glycol, Water; 60 °C; 60 °C → 170 °C; 4 h, reflux; reflux → 200 °C; 200 °C → rt

Experimental Protocols

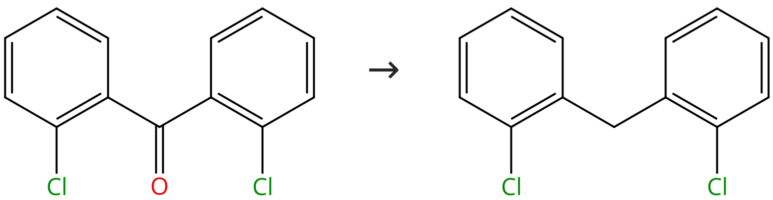
Synthesis and structure of 7- alkoxyalkyl-3-thia-7-azabicyclo [3.3.1]nonan-9-ones and several of their deriva tives

By: Yu, V. K.; et al

Chemistry of Heterocyclic Compounds (New York, NY, United States) (2006), 42(4), 512-519.

Scheme 42 (1 Reaction)

Steps: **1** Yield: **91%**



Suppliers (49)

Suppliers (10)

31-538-CAS-2550609

Steps: **1** Yield: **91%**

1.1 **Reagents:** Hydrazine

1.2 **Reagents:** Potassium hydroxide

Experimental Protocols

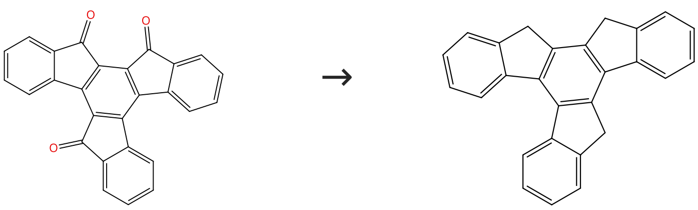
Microwave induced synthesis of hydrazones and Wolff-Kishner reduction of carbonyl compounds

By: Gadhwal, Sunil; et al

Synlett (1999), (10), 1573-1574.

Scheme 43 (1 Reaction)

Steps: **1** Yield: **91%**



Suppliers (2)

Suppliers (3)

31-538-CAS-7571102

Steps: **1** Yield: **91%**

1.1 **Reagents:** Potassium hydroxide, Hydrazine hydrate (1:1)
Solvents: Ethylene glycol diethyl ether; 24 h, 180 °C; 180 °C → 60 °C

1.2 **Reagents:** Hydrochloric acid
Solvents: Water; 0 °C

Experimental Protocols

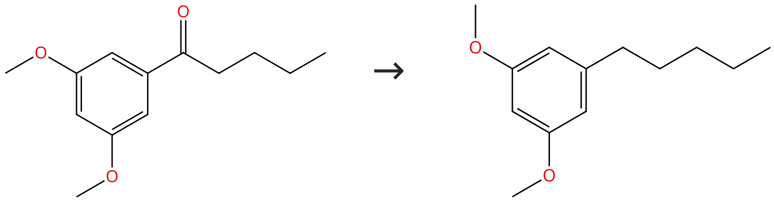
Facile Multistep Synthesis of Isotruxene and Isotru xenone

By: Yang, Jye-Shane; et al

Journal of Organic Chemistry (2009), 74(10), 3974-3977.

Scheme 44 (1 Reaction)

Steps: **1** Yield: **91%**



Suppliers (24)

Suppliers (46)

31-399-CAS-4599064

Steps: **1** Yield: **91%**

1.1 **Reagents:** Hydrazine hydrate (1:1); 2 h, reflux; 3 h, reflux; reflux → rt

Experimental Protocols

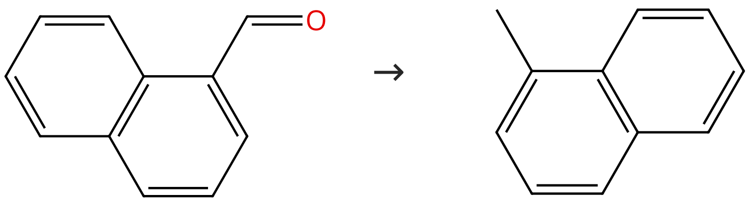
New process for the preparation of 3,5-dihydroxy-1-pentylbenzene

By: Zhang, Long; et al

Research on Chemical Intermediates (2007), 33(6), 535-540.

Scheme 45 (1 Reaction)

Steps: **1** Yield: **91%**



Suppliers (83)

Suppliers (75)

31-538-CAS-12900822

Steps: **1** Yield: **91%**

1.1 **Reagents:** Potassium hydroxide, Hydrazine hydrate (1:1)
Solvents: Diethylene glycol monoethyl ether; 3 min, 200 psi, 200 °C

Experimental Protocols

Rapid Wolff-Kishner reductions in a silicon carbide microreactor

By: Newman, Stephen G.; et al

Green Chemistry (2014), 16(1), 176-180.